Project #1: Modular Data Workflow for Wine Quality Analysis

Course: MSc Capstone / AI Programming Foundations | **Dataset:** UCI Wine Quality Dataset (winequality-red.csv)

Executive Summary: This academic summary report documents a modular, reproducible Python data processing and exploratory analysis workflow examining the physicochemical attributes of red wine. Using **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn**, the pipeline ingests sensor metrics, standardizes column structures, discretizes quality scores into ordinal tiers, computes grouped summary statistics, and generates publication-ready visualizations.

1. Methodology & Data Processing Workflow

The data workflow was developed in a structured Jupyter environment (data_workflow.ipynb) following modular software engineering principles. The workflow executes four sequential stages:

- **Data Ingestion (Task 4):** Ingested raw sensor metrics from winequality-red.csv (1,599 observations, 12 attributes). Implemented semicolon delimiter parsing (sep=";") to correctly parse UCI dataset formatting and validated structural dimensions.
- **Column Standardization Function (Task 5.1):** Executed clean_column_names() to strip whitespace, lowercase text, and convert column headers into standardized snake_case (e.g., fixed acidity -> fixed_acidity). This prevents syntax errors during attribute indexing.
- **Feature Engineering & Discretization Function (Task 5.2):** Executed add_quality_tier() using pd.cut() to bin continuous quality scores (3 to 8) into three ordinal tiers: **Low** (3-4), **Medium** (5-6), and **High** (7-8). This categorization facilitates multi-attribute comparisons across quality tiers.
- **Exploratory Data Analysis Function (Task 6):** Executed compute_wine_summary() to compute aggregated metrics (mean, standard deviation, median) across key chemical properties grouped by quality_tier.

2. Key Analytical Findings & Summary Statistics

Grouped aggregation revealed significant physicochemical differences across wine quality tiers, as summarized in Table 1 below:

Quality Tier	Alcohol (%) Mean +/- Std (Med)	Volatile Acidity (g/dm^3) Mean +/- Std (Med)	Citric Acid (g/dm^3) Mean +/- Std (Med)	Sulphates (g/dm^3) Mean +/- Std (Med)	pH Mean +/- Std (Med)
Low (3-4) ***n = 63***	10.22 +/- 0.92 (10.00)	0.724 +/- 0.248 (0.680)	0.174 +/- 0.207 (0.080)	0.592 +/- 0.224 (0.560)	3.384 +/- 0.175 (3.380)
Medium (5-6) ***n = 1,319***	10.25 +/- 0.97 (10.00)	0.539 +/- 0.168 (0.540)	0.258 +/- 0.188 (0.240)	0.647 +/- 0.167 (0.610)	3.311 +/- 0.152 (3.310)
High (7-8) ***n = 217***	11.52 +/- 1.00 (11.60)	0.406 +/- 0.145 (0.370)	0.376 +/- 0.194 (0.400)	0.743 +/- 0.134 (0.740)	3.289 +/- 0.154 (3.270)

Primary Physicochemical Drivers (Task 7 Visual Insights):

- **Alcohol Concentration (+0.48 correlation with quality):** High-tier wines exhibit a significantly higher average alcohol concentration (11.52% avg, 11.60% median) compared to Medium (10.25%) and Low (10.22%) tiers. The box plot visualization confirms that higher alcohol content consistently aligns with higher expert rating distributions.
- **Volatile Acidity (-0.39 correlation with quality):** Volatile acidity (associated with acetic acid and vinegar flavor notes) demonstrates a steep inverse relationship with quality, declining from 0.724 g/dm^3 in Low tier to 0.406 g/dm^3 in High tier.
- **Citric Acid & Sulphates (+0.23 and +0.25 correlations):** Citric acid (which imparts fresh citrus notes) more than doubles from Low (0.174 g/dm^3) to High (0.376 g/dm^3) tiers. Sulphates (antimicrobial and antioxidant agents) increase steadily from 0.592 to 0.743 g/dm^3, contributing to preservation and sensory balance.

3. Bias & Limitations Analysis

A rigorous evaluation of the data workflow identifies three primary limitations regarding dataset sampling, scope, and target variable distribution:

- **Geographic Sampling Bias:** All samples were collected exclusively from the Vinho Verde region in northern Portugal. The chemical properties reflect localized microclimates, soil conditions, and specific grape varieties (e.g., Vinhao). Consequently, the quantitative thresholds identified here may not generalize to wines produced in other global viticultural regions (e.g., Napa Valley, Bordeaux, or Barossa Valley).
- **Omission of White Wine Metrics:** This workflow focused strictly on red wine samples (winequality-red.csv). White wine varieties (winequality-white.csv) exhibit distinct physicochemical structures, including significantly higher residual sugar levels, different sulfur dioxide dynamics, and distinct acidity profiles.
- **Sensory Panel Subjectivity & Class Imbalance:** Quality scores reflect human sensory evaluations averaged across expert assessors. Scores are heavily concentrated in Medium quality ratings (82.5% of samples in scores 5 and 6), with extreme scores (0-2 and 9-10) completely unrepresented. This target variable compression creates class imbalance that must be addressed via re-sampling or cost-sensitive learning when building predictive models.

4. References (APA 7th Edition)

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